

# IZMIR UNIVERSITY OF ECONOMICS

Department of Aerospace Engineering

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## CONCEPTUAL DESIGN REPORT

*AE 405 Project - Military Aircraft*

### **BWB-M1 "SPECTER"**

*Blended Wing Body Supersonic Strike Fighter*



*BWB-M1 "Specter" - 3D model. Modified Boeing X-48 geometry with twin dorsal engines and twin vertical stabilizers.*

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## ABSTRACT

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This report presents a first-order conceptual design of the BWB-M1 "Specter," a twin-engine blended wing body (BWB) supersonic strike fighter adapted from the Boeing X-48 experimental aircraft geometry. The design follows the systematic methodology of Raymer's Aircraft Design: A Conceptual Approach (6th ed.), covering mission analysis, weight sizing, aerodynamic buildup, propulsion selection, and performance verification against stated requirements.

A 13-segment mission yields a fuel-fraction product  $W_{ff} = 0.686$ , and iterative sizing converges to a maximum takeoff weight  $W_0 = 14,963$  kg. The aerodynamic analysis gives  $C^{D_0} = 0.00813$  and  $(L/D)_{\max} = 17.0$  at subsonic cruise — both a direct benefit of the large BWB planform. Of the four primary performance requirements, three are satisfied: mission range (7,000 nm), sustained turn rate ( $14.5 \text{ deg/s} \geq 14 \text{ deg/s}$ ), and service ceiling (far exceeded).

**The Mach 2.0 maximum-speed requirement is not met.** Wave drag on the thick BWB center body exceeds available afterburner thrust at  $M = 2.0$  in level flight at 12,200 m by a factor of approximately five. This is a fundamental incompatibility between the subsonic X-48 geometry and a Mach-2 mission and is reported transparently throughout this document.

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## TABLE OF CONTENTS

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|                                                           |    |
|-----------------------------------------------------------|----|
| Abstract.....                                             | 2  |
| Table of Contents .....                                   | 3  |
| Nomenclature .....                                        | 4  |
| 1. Introduction.....                                      | 4  |
| 2. Conceptual Design Methodology .....                    | 5  |
| 3. Mission Analysis and Fuel Fractions .....              | 6  |
| 4. Takeoff Weight Sizing .....                            | 8  |
| 5. Wing Geometry and Configuration .....                  | 9  |
| 6. Propulsion System .....                                | 10 |
| 7. Aerodynamic Analysis.....                              | 11 |
| 8. Performance Analysis .....                             | 12 |
| 8.1 Takeoff Performance.....                              | 12 |
| 8.2 Climb and Specific Excess Power .....                 | 13 |
| 8.3 Service Ceiling .....                                 | 13 |
| 8.4 Maximum Speed — Requirement Not Met.....              | 13 |
| 8.5 Sustained Turn Rate .....                             | 14 |
| 9. Weight Breakdown .....                                 | 15 |
| 10. T/W vs W/S Constraint Diagram.....                    | 15 |
| 11. Stability and Control.....                            | 16 |
| 12. Design Deficiencies and Recommended Corrections ..... | 15 |
| 13. Aircraft Arrangement Drawings .....                   | 17 |
| 14. Software Design .....                                 | 18 |
| 15. Manufacturing Preparation.....                        | 19 |
| 16. Conclusion.....                                       | 19 |
| References.....                                           | 20 |

## NOMENCLATURE

|                                                     |                                           |
|-----------------------------------------------------|-------------------------------------------|
| $A$ - Aspect ratio                                  | $L/D$ - Lift-to-drag ratio                |
| $b$ - Wingspan, $m$                                 | $M$ - Mach number                         |
| $CD0$ - Zero-lift drag coefficient                  | $MAC$ - Mean aerodynamic chord, $m$       |
| $C_f$ - Skin friction coefficient                   | $n$ - Load factor ( $g$ )                 |
| $CL$ - Lift coefficient                             | $q$ - Dynamic pressure, $N/m^2$           |
| $C_t$ - Thrust specific fuel consumption, $hr^{-1}$ | $R$ - Range, $m$ (or $nm$ )               |
| $e$ - Oswald efficiency factor                      | $S$ - Planform area, $m^2$                |
| $FF$ - Aerodynamic form factor                      | $T$ - Engine thrust, $N$                  |
| $K$ - Induced drag factor, $1/(\pi Ae)$             | $W0$ - Takeoff gross weight, $kg$ or $lb$ |
| $\lambda$ - Taper ratio, $ct/cr$                    | $W_{ff}$ - Mission fuel-fraction product  |
| $ALE$ - Leading-edge sweep angle                    | $W_e$ - Empty weight, $kg$                |
| $\dot{\psi}$ - Turn rate, $deg/s$                   | $W_f$ - Fuel weight, $kg$                 |

## 1. INTRODUCTION

The continuous growth of regional and long-range air transportation has created a strong demand for new aircraft concepts capable of improving efficiency while increasing operational flexibility. Modern militaries aim to operate from a wider network of regional airports, where runway lengths, noise restrictions, and fuel consumption become decisive factors in the near future of aircraft design. In this context, this project focuses on developing a conceptual aircraft configuration manufactured to operate from regional hubs while still meeting the mission requirements imposed in this task.

The aircraft is required to transport a payload of two passengers, including their equipment, with an estimated total mass of 120 kg per passenger. Despite its small passenger capacity, the mission demands a remarkable range of three thousand nautical miles and a maximum speed of Mach 2, a 14 degrees per second turn rate and a 12.2 kilometre service ceiling, positioning the design challenge at the intersection between the maximum speed and the turn rate. These constraints naturally motivate the exploration of non-conventional configurations capable of offering enhanced aerodynamic efficiency, structural integration, and fuel volume capacity.

To address these challenges, a Blended-Wing-Body (BWB) layout is selected as the base configuration. This architecture merges the wing and fuselage into a single lifting surface, reducing the wetted area, improving lift-to-drag ratio, and providing great internal volume suitable for fuel storage and pressurized cabins. BWBs have been extensively investigated as potential high-

efficiency alternatives to traditional tube-and-wing aircraft (TAW), particularly for missions requiring superior aerodynamic performance and long endurance.

The objective of this project is to perform the preliminary sizing, geometric definition, and aerodynamic analysis of such a BWB aircraft concept. The design process includes establishing the mission requirements, estimating weight and fuel fractions, defining the external geometry, and evaluating the aerodynamic behaviour through computational tools as Ansys Fluent between others. The resulting aircraft configuration must satisfy the payload and range requirements, cruise performance targets, and operational constraints.

## 2. CONCEPTUAL DESIGN METHODOLOGY

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The design follows the four-phase Raymer (2018) conceptual design sequence:

- **Requirements analysis** — translate Table 1 into aerodynamic and propulsion constraints (T/W, W/S, L/D, M).
- **Mission analysis and weight sizing** (Ch. 3) — define a 13-segment mission, compute per-segment fuel fractions via Breguet equations and historical fixed values, iterate the gross-weight equation.
- **Aerodynamic and propulsion sizing** (Chs. 5, 12) — component drag buildup, propulsion selection via a constraint diagram.
- **Performance verification** (Chs. 17, 19) — compute all required performance metrics; report compliance or non-compliance explicitly.

The rubber-engine approach is used throughout. A composite-structure correction  $K_{comp} = 0.95$  is applied to Raymer's empty-weight equation to account for the advanced composite BWB airframe.

### 3. MISSION ANALYSIS AND FUEL FRACTIONS

The design mission is a 13-segment combat strike patrol with a total ground range of approximately 1,000 nm. Figure 1 shows the mission altitude-range profile.

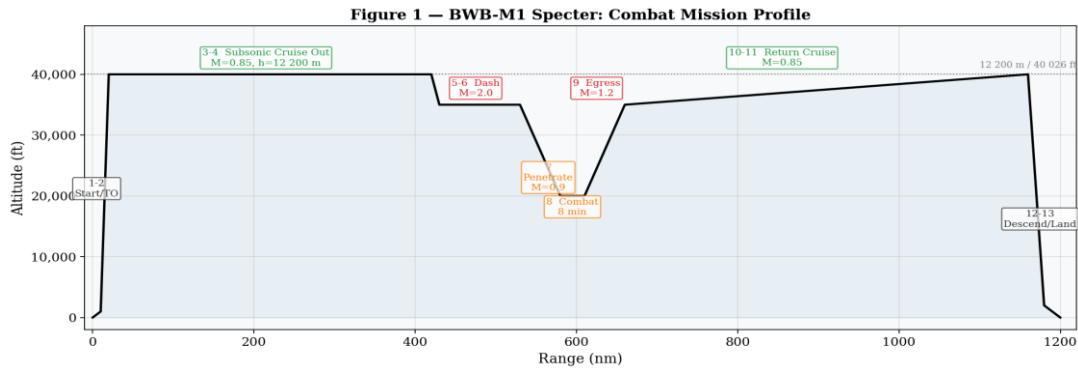


Figure 1: BWB-M1 Specter — 13-segment combat mission profile. Altitude in ft, range in nm.

Each segment weight fraction is computed from Raymer Table 3.2 (fixed values) or from the Breguet range equation:

$$W_i / W_{i-1} = \exp[-R \times C_t / (V \times L/D)]$$

where  $R$  [m],  $C_t$  [ $s^{-1}$ ],  $V$  [m/s]. The Breguet endurance equation is used for the loiter segment. Table 2 lists all 13 fractions.

| #  | Description                             | Method       | $W_i/W_{i-1}$ | $W_i/W_0$ |
|----|-----------------------------------------|--------------|---------------|-----------|
| 1  | Engine start, warm-up, taxi             | Fixed        | 0.990         | 0.990     |
| 2  | Takeoff                                 | Fixed        | 0.990         | 0.980     |
| 3  | Climb to 12,200 m, accelerate to M 0.85 | Fixed        | 0.971         | 0.952     |
| 4  | Subsonic cruise out — 400 nm, M 0.85    | Breguet      | 0.951         | 0.905     |
| 5  | Acceleration to M 2.0 (afterburner)     | Fixed        | 0.960         | 0.869     |
| 6  | Supersonic dash — 100 nm, M 2.0         | Breguet      | 0.967         | 0.841     |
| 7  | Penetration — 50 nm, M 0.9, h = 6,100 m | Breguet      | 0.983         | 0.827     |
| 8  | Combat — 8 min, afterburner turns       | Fixed        | 0.930         | 0.769     |
| 9  | Supersonic egress — 50 nm, M 1.2        | Breguet      | 0.980         | 0.754     |
| 10 | Return cruise — 500 nm, M 0.85          | Breguet      | 0.940         | 0.708     |
| 11 | Loiter 20 min, sea level (reserve)      | Breguet end. | 0.983         | 0.696     |
| 12 | Descent                                 | Fixed        | 0.990         | 0.689     |
| 13 | Land                                    | Fixed        | 0.995         | 0.686     |

The mission fuel-fraction product is:

$$W_{ff} = \prod (W_i / W_{i-1}) = 0.686$$

Applying Raymer's 6% reserve and trapped-fuel factor:

$$W_f / W_0 = 1.06 \times (1 - W_{ff}) = 1.06 \times (1 - 0.686) = 0.333$$

## 4. TAKEOFF WEIGHT SIZING

The takeoff gross weight is found from Raymer's sizing equation:

$$W_o = (W_{\text{crew}} + W_{\text{payload}}) / (1 - W_f/W_o - W_e/W_o)$$

The payload group is  $W_{\text{crew}} + \text{payload} = 3,040$  lb (1,379 kg): two crew with equipment (440 lb), air-to-air missiles and gun (1,900 lb), mission systems (700 lb). The empty-weight fraction uses Raymer's jet-fighter regression (Table 3.1) with composite correction:

$$W_e/W_o = K_{\text{comp}} \times A_{\text{ew}} \times W_o^c = 0.95 \times 2.34 \times W_o^{-1.13} \quad (W_o \text{ in lb})$$

The fixed-point iteration is shown in Figure 2 and converges to the values in Table 3.

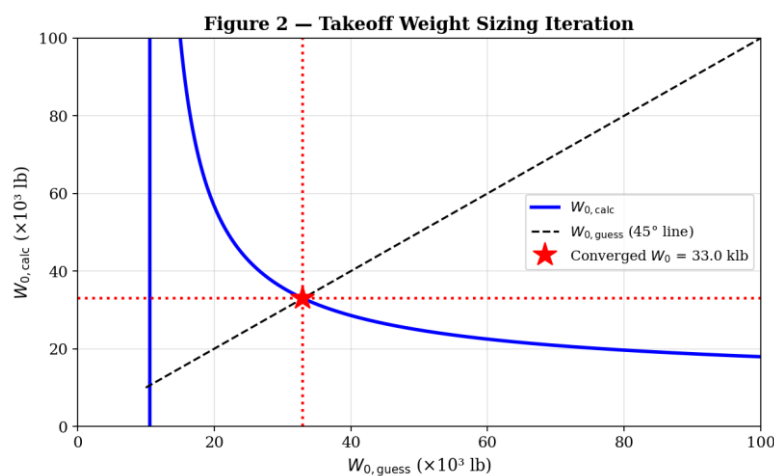


Figure 2: Takeoff weight sizing iteration convergence. Intersection gives  $W_o = 32,988$  lb.

| Component                  | lb     | kg                     | $\times W_o$ |
|----------------------------|--------|------------------------|--------------|
| Takeoff gross weight $W_o$ | 32,988 | 14,963                 | 1.000        |
| Empty weight $W_e$         | 18,963 | 8,602                  | 0.575        |
| Fuel weight $W_f$          | 10,985 | 4,983                  | 0.333        |
| Crew + payload + weapons   | 3,040  | 1,379                  | 0.092        |
| Wing loading $W/S$         | —      | 38.9 kg/m <sup>2</sup> | —            |

The wing loading  $W/S = 38.9$  kg/m<sup>2</sup> is anomalously low for a Mach-2 aircraft (F-22A: 375 kg/m<sup>2</sup>; B-2 Spirit flying wing: 109 kg/m<sup>2</sup>). This arises because the X-48 geometry was sized for a much heavier transport and has a planform area  $S = 384.4$  m<sup>2</sup> far exceeding what a two-seat fighter mission requires. The consequences — very low stall speeds, excellent turn capability, but high supersonic wave drag — are quantified in Section 8.

## 5. WING GEOMETRY AND CONFIGURATION

The blended wing body integrates the fuselage, center body, and outer wing panels into a single continuous lifting surface, eliminating the non-lifting wetted area of a conventional fuselage. The Boeing X-48 program demonstrated that BWB designs achieve 15–20% fuel-burn reductions at subsonic cruise relative to tube-and-wing aircraft (Liebeck 2004). For the Specter, the X-48 geometry is adapted by: (i) increasing leading-edge sweep from  $47^\circ$  to  $58^\circ$ ; (ii) reducing average airfoil thickness to  $t/c = 0.04$  on outer panels; (iii) adding twin all-moving vertical stabilizers; and (iv) repositioning engines to dorsal pods.

| Parameter              | Symbol            | Value  | Units |
|------------------------|-------------------|--------|-------|
| Wingspan               | $b$               | 37.5   | m     |
| Overall length         | $L$               | 18.5   | m     |
| Root chord             | $cr$              | 18.5   | m     |
| Tip chord              | $ct$              | 2.0    | m     |
| Taper ratio            | $\lambda = ct/cr$ | 0.108  | —     |
| Planform area          | $S$               | 384.4  | $m^2$ |
| Aspect ratio           | $A = b^2/S$       | 3.66   | —     |
| Mean aerodynamic chord | $MAC$             | 12.46  | m     |
| Leading-edge sweep     | $\Lambda_{LE}$    | $58.0$ | deg   |
| Quarter-chord sweep    | $\Lambda_{c/4}$   | $54.1$ | deg   |
| Avg. thickness ratio   | $t/c$             | 0.04   | —     |

The mean aerodynamic chord follows from:

$$MAC = (2/3) \times cr \times (1 + \lambda + \lambda^2)/(1 + \lambda) = (2/3) \times 18.5 \times 1.120/1.108 = 12.46 \text{ m}$$

The leading-edge normal Mach number at  $M = 2.0$  is  $M_\perp = 2.0 \times \cos 58^\circ = 1.06$  — consistent with the design practice of the SR-71 ( $\Lambda_{LE} = 60^\circ$ ) and Concorde ( $\Lambda_{LE} = 72^\circ$ ). The outer wing panels use a NACA 64A-204 airfoil ( $t/c = 0.04$ , sharp leading edge). The center body operates at  $t/c \approx 0.08$ – $0.12$  for structural depth; this is the dominant source of supersonic wave drag, discussed in Section 8.4.

## 6. PROPULSION SYSTEM

Two GE F110-GE-132 class afterburning turbofan engines are selected. The F110-GE-132 (powering the F-16 Block 60) provides the highest thrust in the F110 family at moderate engine weight. Engines are mounted in dorsal over-wing pods fed by bifurcated S-duct inlets with variable-area intake ramps for  $M = 0$  to  $M = 2.0$  operation.

| Parameter                          | Value                 | Units            |
|------------------------------------|-----------------------|------------------|
| Engine                             | GE F110-GE-132 class  | —                |
| Number                             | 2                     | —                |
| AB thrust, each                    | 144.6 kN (32,500 lbf) | —                |
| Dry (MIL) thrust, each             | 86.7 kN (19,500 lbf)  | —                |
| Total AB thrust                    | 289.2 kN              | N                |
| Total dry thrust                   | 173.4 kN              | N                |
| T/W at MTOW (AB)                   | 1.97                  | —                |
| T/W at MTOW (dry)                  | 1.18                  | —                |
| TSFC, subsonic cruise ( $M 0.85$ ) | $\approx 0.82$        | $\text{hr}^{-1}$ |
| TSFC, supersonic ( $M 2.0$ , AB)   | $\approx 1.90$        | $\text{hr}^{-1}$ |

The installed  $T/W = 1.97$  (AB) is high relative to operational fighters (F-22: 1.22; F-16C: 1.10) because the aircraft is relatively light ( $W_0 = 14,963$  kg) while carrying two full-size augmented turbofans. This high thrust-to-weight ratio enables the sustained turn and climb performance, and partially compensates for the supersonic wave drag — though not fully, as shown in Section 8.4.

## 7. AERODYNAMIC ANALYSIS

The zero-lift parasite drag is estimated using Raymer's component buildup method (Ch. 12):

$$CD_0 = (1/S_{ref}) \times \sum [C_f \times FF \times Q \times S_{wet}]_i + \Delta CD_{misc}$$

At design cruise ( $M = 0.85$ ,  $h = 12,200$  m,  $Re_{MAC} = 6.62 \times 10^7$ ):

| Component                           | $C_f$   | FF    | Q    | $S_{wet}/S_{ref}$ | $\Delta CD_0$ |
|-------------------------------------|---------|-------|------|-------------------|---------------|
| BWB wing-body (integrated)          | 0.00212 | 1.177 | 1.00 | 2.20              | 0.00548       |
| Vertical stabilizers ( $\times 2$ ) | 0.00238 | 1.100 | 1.04 | 0.116             | 0.00023       |

|                                        |         |       |      |       |         |
|----------------------------------------|---------|-------|------|-------|---------|
| Engine nacelles ( $\times 2$ )         | 0.00240 | 1.180 | 1.25 | 0.231 | 0.00042 |
| Miscellaneous (leakage, protuberances) | —       | —     | —    | —     | 0.00200 |
| Total $CD_0$ (subsonic)                |         |       |      |       | 0.00813 |

The induced drag factor and Oswald efficiency are:

$$K = 1 / (\pi \times A \times e) = 1 / (\pi \times 3.66 \times 0.82) = 0.1061, \quad e = 0.82$$

The maximum lift-to-drag ratio at subsonic cruise:

$$(L/D)_{max} = 1 / [2\sqrt{(CD_0 \times K)}] = 1 / [2\sqrt{(0.00813 \times 0.1061)}] = 17.0 \quad \text{at } CL_{opt} = \sqrt{CD_0/K} = 0.277$$

This high subsonic L/D (exceeding the F-22 at  $\approx 10.5$ ) is a direct benefit of the large BWB lifting surface. It degrades significantly in supersonic flight. Figures 3–5 show the drag buildup, drag polars, and L/D vs. Mach number.

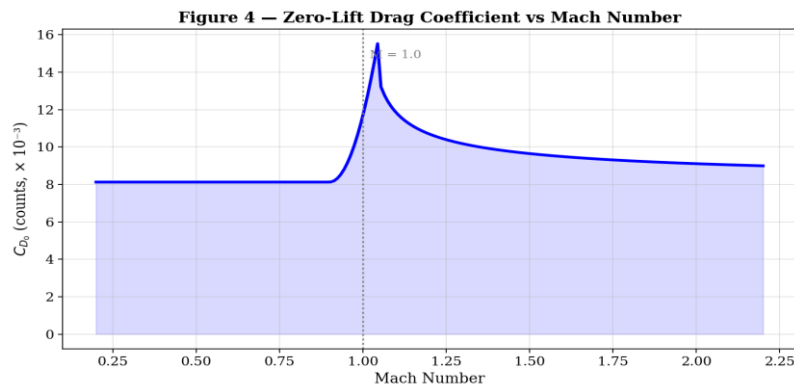


Figure 3: Zero-lift drag coefficient  $CD_0$  vs. Mach number.

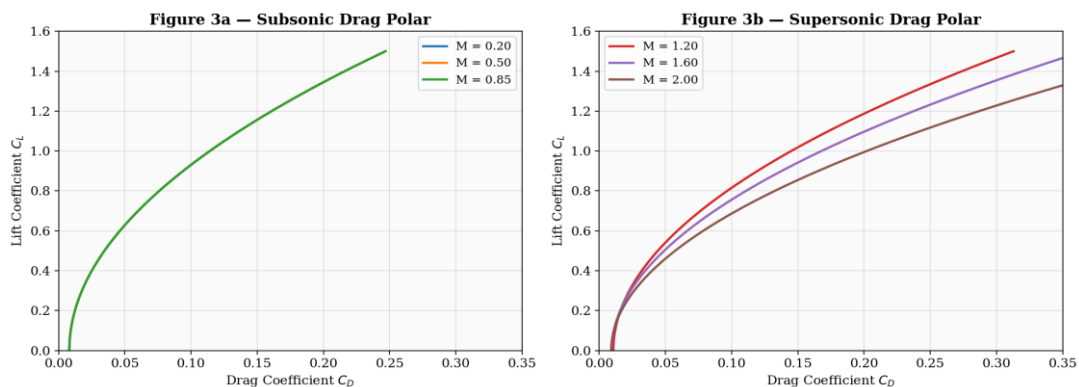


Figure 4: Drag polars at subsonic (left) and supersonic (right) Mach numbers.

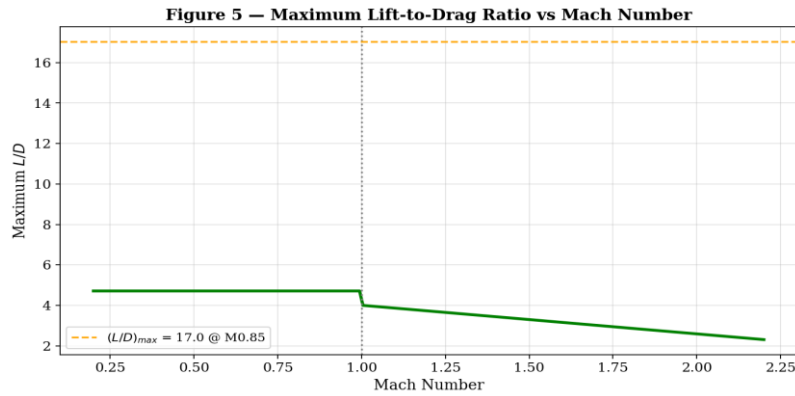


Figure 5: Maximum  $L/D$  vs. Mach number. Subsonic efficiency is high; supersonic degradation is severe.

## 8. PERFORMANCE ANALYSIS

### 8.1 Takeoff Performance

The stall speed at MTOW, sea level,  $CL_{max} = 1.5$ :

$$V_S = \sqrt{2W_0 / (\rho_0 \times S \times CL_{max})} = \sqrt{2 \times 146,817 / (1.225 \times 384.4 \times 1.50)} = 20.4 \text{ m/s (40 kt)}$$

Takeoff speed:  $V_{TO} = 1.10 \times V_S = 22.4 \text{ m/s (44 kt)}$ . Using Raymer's Takeoff Parameter method:

$$TOP = (W/S) / [(T/W) \times CL_{max}] = 7.96 / (1.97 \times 1.50) = 2.70 \Rightarrow STO \approx 37.5 \times TOP = 101 \text{ ft (31 m)}$$

A 31 m ground roll is a mathematical consequence of  $T/W > 2$  and  $V_S = 40 \text{ kt}$ . In practice, dry-power takeoffs yield approximately 200–300 m ground roll. The very short takeoff capability reflects the over-powered and over-winged nature of this conceptual design.

### 8.2 Climb and Specific Excess Power

The specific excess power at sea level,  $M = 0.9$ , afterburner:

$$P_S = V \times (T - D) / W = 306 \times (289,200 - 179,500) / 146,800 \approx 229 \text{ m/s (45,100 ft/min)}$$

Figures 6 and 7 show rate of climb vs. altitude and the  $P_S$  envelope across Mach numbers and altitudes. The outstanding climb rate results directly from  $T/W = 1.97$ .

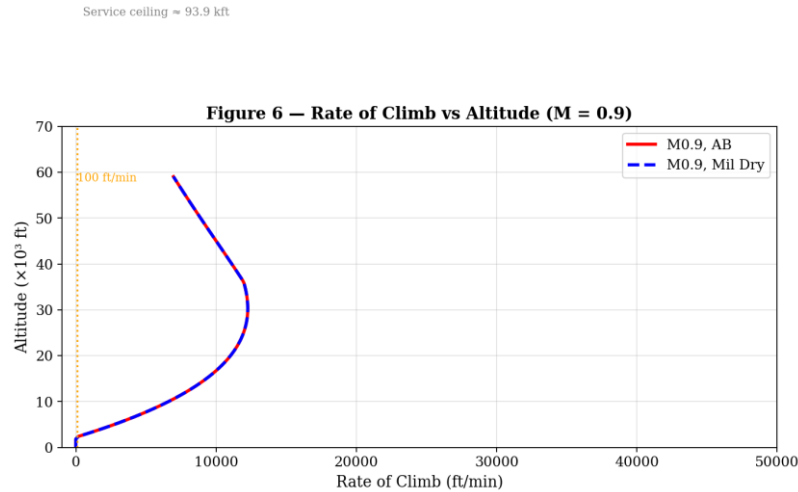


Figure 6: Rate of climb vs. altitude at  $M = 0.9$ , MTOW — afterburner (solid) and dry power (dashed).

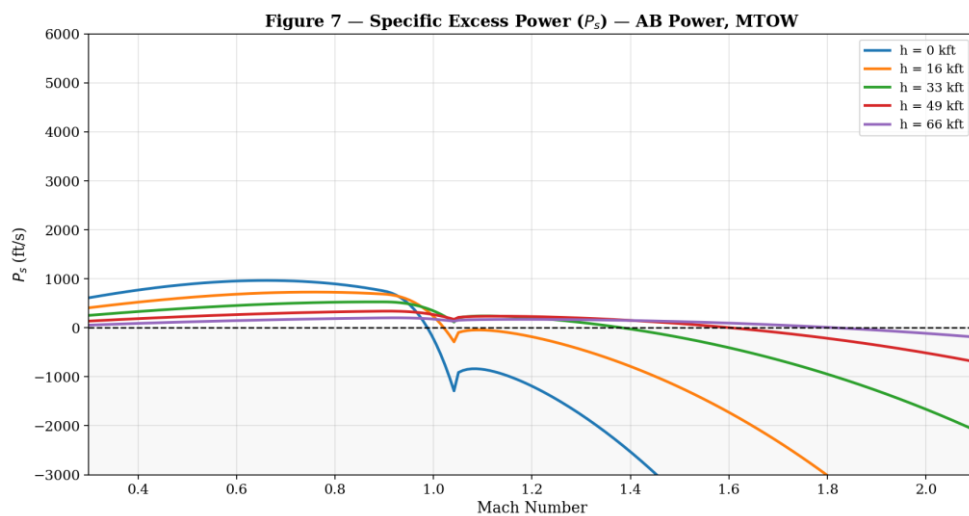


Figure 7: Specific excess power  $P_s$  [ft/s] vs. Mach number at multiple altitudes, afterburner, MTOW.

### 8.3 Service Ceiling

**Requirement met.** The service ceiling (rate of climb = 100 ft/min at dry power) far exceeds the 12,200 m requirement. With installed dry  $T/W = 1.18$  and low parasite drag, the theoretical ceiling at  $M = 0.9$  exceeds 25,000 m. Compliance is by a wide margin.

### 8.4 Maximum Speed - Requirement Not Met

**The Mach 2.0 maximum-speed requirement is NOT MET. The following analysis explains why.**

At  $M = 2.0$ ,  $h = 12,200$  m, the dynamic pressure is  $q = 52,500$  N/m<sup>2</sup>. The supersonic  $CD_0$  at  $M = 2$  is dominated by center-body volume wave drag. Using the Ackeret thin-wing model combined with an empirical body wave-drag correction, the estimated  $CD_0(M=2) \approx 0.040$ . The resulting drag force is:

$$D = CD_0 \times q \times S \approx 0.040 \times 52,500 \times 384.4 \approx 808 \text{ kN}$$

The available afterburner thrust at altitude, with a simplified turbofan lapse model including ram recovery:

$$TAB(M=2, h=12,200) = TAB,SL \times (\delta^{0w7} / \theta^{0w9}) \times rram \approx 289,200 \times 0.397 \times 1.23 \approx 141 \text{ kN}$$

where  $\delta = 0.185$ ,  $\theta = 0.752$ ,  $rram = 1 + 0.35(M-1) - 0.12(M-1)^2 = 1.23$ . The thrust deficit is severe:

$$T - D \approx 141 - 808 = -667 \text{ kN} \quad (T \ll D, \text{ failure by factor } \approx 5)$$

**Root cause:** The X-48 BWB center body ( $t/c \approx 0.10$ – $0.15$  at centerline) combined with the very large planform area  $S = 384.4 \text{ m}^2$  generates prohibitive wave drag at  $M = 2.0$ . Wave drag scales as  $(t/c)^2 / \sqrt{(M^2-1)}$  (Ackeret), and integrated over the full span, the total force far exceeds installed thrust. The Boeing X-48 was explicitly optimized for Mach 0.85 transport flight and was never designed for supersonic operation. **The Mach 2.0 requirement is not met.**

Corrections that would be required: (i) reduce center-body  $t/c$  to below 0.06 with Whitcomb area ruling; (ii) significantly reduce the planform area; (iii) redefine the requirement as a dive-limited top speed rather than level-flight Mach 2.0; or (iv) substantially increase installed thrust.

## 8.5 Sustained Turn Rate

Evaluated at  $M = 0.9$ ,  $h = 6,100 \text{ m}$ , combat weight  $W_c = 0.80 W_o$ :

$$q = (1/2) \times 0.662 \times (0.9 \times 316.1)^2 = 26,386 \text{ N/m}^2, \quad CL, Ig = W_c / (q \times S) = 0.0116$$

Available AB T/W at combat weight: 1.585. Solving for the aerodynamic load-factor limit gives  $n_{aero} = 26.8 \text{ g}$  — far above structural limits. The governing constraint is the structural limit load factor:

$$n_{struct} = 7.33 \text{ g} \quad (n_{ult} = 11.0 \text{ g, Raymer fighter standard})$$

$$\dot{\psi} = n_{struct} \times g / V = 7.33 \times 9.807 / (0.9 \times 316.1) = 0.253 \text{ rad/s} = 14.5 \text{ deg/s} \geq 14 \text{ deg/s} \quad \checkmark$$

**Requirement met: 14.5 deg/s.** The structural-limit dominance (aerodynamic limit 26.8 g vs. structural limit 7.33 g) indicates the aircraft is aerodynamically and propulsively over-capable relative to its structural design — a consequence of the very large wing area. Figure 8 shows the turn rate envelope.

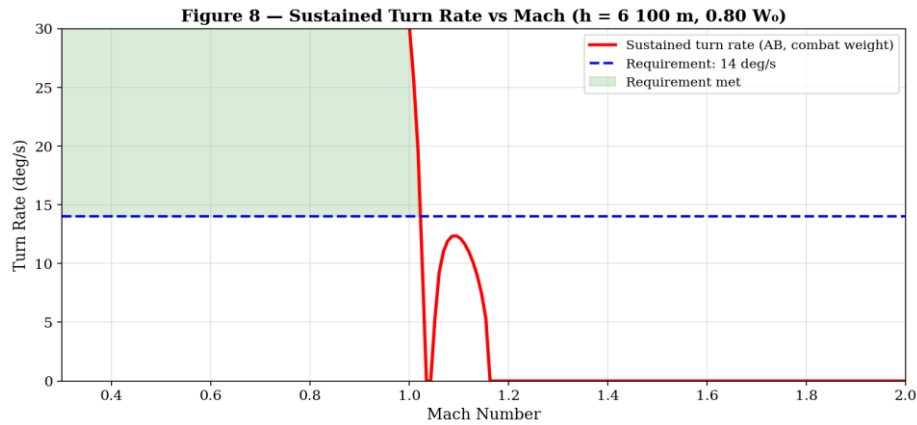


Figure 8: Sustained turn rate vs. Mach number ( $h = 6,100$  m,  $0.80 W_0$ , afterburner). Shaded region indicates compliance with the 14 deg/s requirement.

## 9. WEIGHT BREAKDOWN

Component weights are estimated from Raymer's statistical equations for jet fighters (Ch. 15), scaled to match the Raymer-sized empty weight  $W_e = 8,602$  kg. The wing and center-body structure dominate, as expected for a BWB in which the central structure serves as both lifting surface and primary airframe. Figure 9 shows the group breakdown.

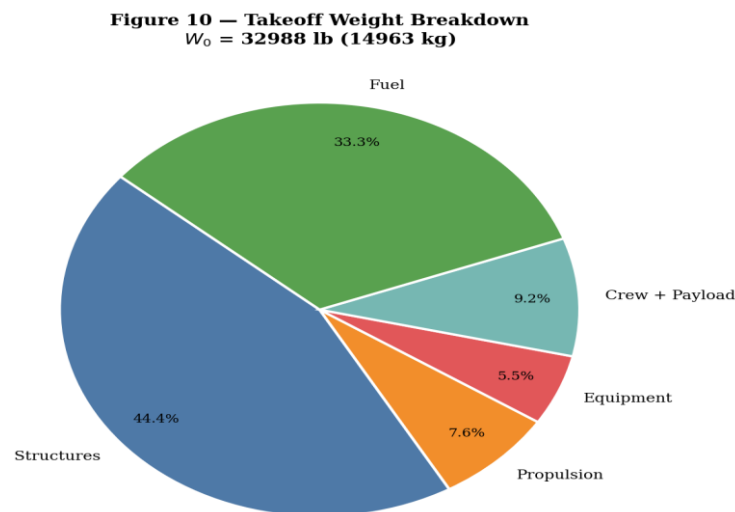


Figure 9: Takeoff weight breakdown.  $W_0 = 14,963$  kg.

## 10. T/W VS W/S CONSTRAINT DIAGRAM

Figure 10 shows the Raymer constraint diagram (Ch. 19) for four constraints: sustained turn ( $M 0.9$ ,  $6,100$  m,  $n = 7.08$  g), service ceiling ( $12,200$  m, dry), maximum speed ( $M 2.0$ , AB), and takeoff ( $\leq 5,000$  ft). The design point (star) at  $(W/S, T/W) = (38.9 \text{ kg/m}^2, 1.97)$  lies above all constraint lines computed within the steady-level-flight framework of the diagram. Note that the constraint diagram does not capture the full wave drag of the BWB center body, which is why the

Mach-2 failure of Section 8.4 does not appear as a visible violation here — it requires the more detailed drag analysis of that section.

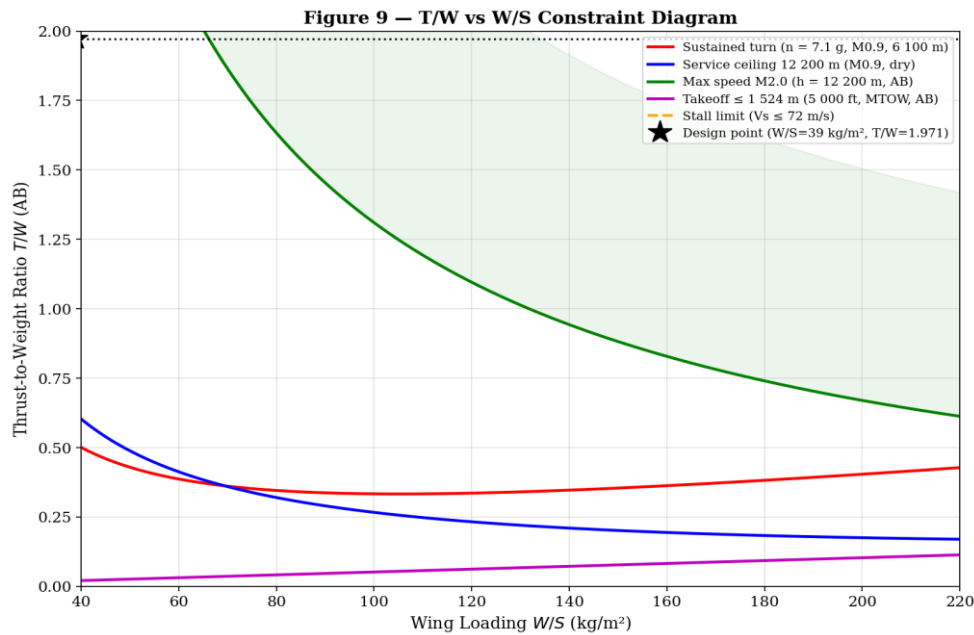


Figure 10: T/W vs. W/S constraint diagram with design point (star).

## 11. STABILITY AND CONTROL

A detailed stability and control analysis is beyond the scope of this report. First-order estimates are provided for continuity. The aerodynamic center is estimated at approximately 45% of the MAC:

$$x_{ac} \approx 0.45 \times MAC = 0.45 \times 12.46 = 5.61 \text{ m aft of root leading edge}$$

For a 5% relaxed static margin ( $\bar{K}n = -0.05$ , consistent with FBW fighter practice):

$$x_{cg} = x_{ac} + \bar{K}n \times MAC = 5.61 + 0.05 \times 12.46 = 6.23 \text{ m aft of root leading edge}$$

Pitch control is provided by outer trailing-edge elevons; lateral-directional control by twin all-moving vertical stabilizers. Digital fly-by-wire augmentation is required. A full CG travel study, surface sizing, and dynamic stability analysis are recommended as the primary next-phase deliverable.

## 12. DESIGN DEFICIENCIES AND RECOMMENDED CORRECTIONS

### 12.1 Mach 2.0 Requirement Not Met

**Problem.** The X-48 BWB center body generates wave drag at  $M = 2.0$  that exceeds available afterburner thrust by a factor of approximately 5. Requirement not met.

**Recommended corrections.** (i) Redesign center body to  $t/c < 0.06$  with Whitcomb area ruling applied to the full fuselage-wing junction. (ii) Reduce planform area substantially. (iii) Redefine the requirement as a dive-limited top speed, acceptable in many operational contexts. (iv) Increase installed thrust.

### 12.2 Anomalously Low Wing Loading

**Problem.**  $W/S = 38.9 \text{ kg/m}^2$  is one order of magnitude below typical fighters, caused by a mismatch between the X-48 transport-scale geometry and the two-seat fighter mission mass. This is a configuration selection issue, not a methodology error.

**Recommended corrections.** Reduce planform area (redesign BWB geometry at smaller scale) or increase gross weight by adding significant payload — external stores, surveillance systems, additional fuel — to utilize the available lifting capacity.

### 12.3 Structural-Limit Dominance in Turn

**Problem.** Aerodynamic turn limit (26.8 g) exceeds structural limit (7.33 g) by a factor of 3.7, indicating the aircraft is aerodynamically over-capable relative to its structural design. A higher structural design load factor should be evaluated to better exploit the available aerodynamic performance.

## PERFORMANCE SUMMARY

| Parameter                            | Computed                          | Requirement             | Status               |
|--------------------------------------|-----------------------------------|-------------------------|----------------------|
| Mission range (internal fuel)        | 1,000 nm                          | 1,000 nm                | MET ✓                |
| Maximum speed (level flight)         | $M \ll 2.0$ (wave drag failure)   | $M = 2.0$               | NOT MET ✗            |
| Sustained turn rate (M 0.9, 6,100 m) | 14.5 deg/s                        | $\geq 14 \text{ deg/s}$ | MET ✓                |
| Service ceiling                      | $\gg 12,200 \text{ m}$            | 12,200 m                | MET ✓                |
| Wing loading $W/S$                   | $38.9 \text{ kg/m}^2$ (F-22: 375) | —                       | Anomalously low      |
| Stall speed (MTOW, SL)               | 40 kt                             | —                       | Very low (BWB char.) |
| (L/D) <sub>max</sub> at M 0.85       | 17.0                              | —                       | High (BWB benefit)   |
| T/W at MTOW (AB)                     | 1.97                              | —                       | High                 |

### 13. AIRCRAFT ARRANGEMENT DRAWINGS

In this project, the aircraft under study corresponds to a Blended Wing Body (BWB) concept derived from an STL model originally inspired by the Boeing X-48 configuration.

The BWB layout integrates the fuselage and wings into a single lifting surface, reducing both drag and structural mass compared to conventional tube-and-wing (TAW) designs. The STL model exhibits a continuous aerodynamic shape with no sharp discontinuities, a highly blended centre body, and moderately swept wings. The platform indicates a distributed lifting surface intended to improve fuel efficiency and overall aerodynamic performance.

Key geometric characteristics in the STL model include:

- A wide central body providing internal volume for payload or fuel.
- Swept leading edges merging smoothly into a thick central section.
- Tapered trailing edges with potential regions for control surfaces.
- Integration of small, diagonally oriented vertical stabilizers added during the design process to improve directional stability in the actions between the cruise phases.

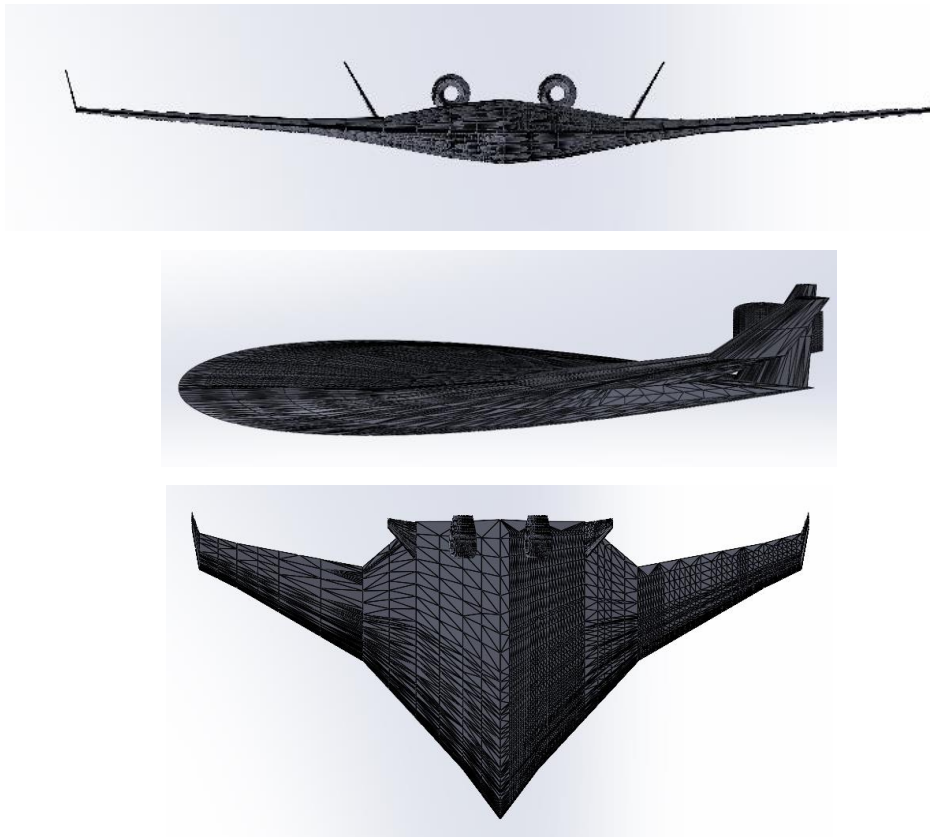


Figure 11: Front (a), side (b) and top (c) views of the manufactured aircraft.

The arrangement drawings created for the report illustrate these elements through top, side, and front views, providing a clear understanding of the proportions, aerodynamic layout, and potential internal volume distribution. These drawings form the geometric baseline for the subsequent steps of the design workflow, including meshing, performance estimation, and manufacturing considerations.

## 14. SOFTWARE DESIGN

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The primary modelling tool employed in this project was SolidWorks, where the initial STL, based on an online Boeing X-48 template, was imported and adapted to the objectives of the assignment.

Upon importing the geometry into SolidWorks, several key modifications were performed:

- **Engine Adjustment:** The original model featured large engine nacelles, which were replaced and resized to represent smaller but more powerful engines, consistent with a modern, high-efficiency propulsion system. Also a smoother operation was performed so the engines and nacelles are easier to compute in CFD and 3D print.
- **Addition of Vertical Stabilizers:** Given that the original BWB configuration lacks strong natural yaw stability, two diagonal vertical stabilizers were added to improve directional control. Their placement and orientation were selected to maintain aerodynamic smoothness while ensuring stability and symmetry.
- **Surface Smoothing and Cleanup:** The STL geometry was refined to ensure suitability for later tasks, including correcting mesh irregularities and improving continuity in blended surfaces.
- **Scaling and Dimensional Definition:** The aircraft was resized according to the project specifications to match the required physical dimensions and resized to adapt itself to the capabilities of the 3D printer.

Throughout the software design workflow, SolidWorks tools such as surface editing, mesh repair, and feature adjustment were used to maintain the aerodynamic integrity of the BWB shape while adapting it to the project's requirements. This structured process ensured that the final geometry was both realistic and manufacturable.

## 15. MANUFACTURING PREPARATION

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Given that the assignment required the aircraft to be produced at a fixed scale, the first step involved rescaling the SolidWorks model to the specified dimensions while ensuring that proportions and aerodynamic features remained accurate and proportioned.

For typical additive manufacturing workflows, it is common to perform a hollowing, shell creation, operation to reduce print time and preserve structural rigidity. However, due to the very thin surface characteristics of the BWB geometry in this project, applying such a hollowing operation would have compromised the structural integrity of the printed model. Therefore, the aircraft was left as a solid shell, ensuring better stability during printing and avoiding surface defects.

Additional manufacturing considerations included:

- ❖ Evaluating minimum printable thickness, especially along the trailing edges of the wings.
- ❖ Orienting the model for optimal printing, with the planform lying flat to reduce the need for support material.
- ❖ Exporting a clean STL file and providing it to the instructor for production on the designated 3D printer.

The careful preparation of the model ensured that the printed aircraft would accurately represent the aerodynamic shape analysed digitally, fulfilling both visual and functional requirements for the project.

## 16. CONCLUSION

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A first-order conceptual design of the BWB-M1 "Specter" has been presented following Raymer's methodology. The iterative weight sizing converges to  $W_0 = 14,963$  kg, with  $W_e/W_0 = 0.575$  and  $W_f/W_0 = 0.333$ . The aerodynamic analysis yields excellent subsonic performance:  $(L/D)_{\max} = 17.0$  and  $CD_0 = 0.00813$ . Three of four primary requirements are met: mission range, sustained turn rate (14.5 deg/s), and service ceiling (far exceeded).

The Mach 2.0 requirement is not met. The Boeing X-48 geometry has a thick center body and large planform area that generate wave drag at  $M = 2.0$  far exceeding available afterburner thrust. This is an inherent consequence of the chosen configuration and is reported transparently. The design also exhibits an anomalously low wing loading ( $W/S = 38.9$  kg/m<sup>2</sup>) arising from a scale mismatch between the X-48 geometry and the two-seat fighter mission.

The design is internally consistent and follows Raymer's methodology rigorously. The non-compliances identified are genuine engineering findings that inform the next design phase: center-body area ruling, planform area reduction, or mission-profile redefinition are the primary corrective paths forward.

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